An Extended Rectangular MinRank Attack against UOV and Its Variants

Toshihiro Suzuki[1,3], Hiroki Furue[2], Takuma Ito[3], Shuhei Nakamura[4], and Shigenori Uchiyama[1]

[1] Tokyo Metropolitan University, Tokyo, Japan
[2] NTT Social Informatics Laboratories, Tokyo, Japan
[3] National Institute of Information and Communications Technology, Tokyo, Japan
[4] Ibaraki University, Ibaraki, Japan

Abstract. Multivariate public key cryptography (MPKC) is considered a promising candidate for post-quantum cryptography, with its security relying on the hardness of solving systems of multivariate quadratic equations. Among MPKC schemes, the unbalanced oil and vinegar (UOV) and its variants have been actively studied. Pébereau and Luyten showed that the Kipnis–Shamir attack and the singular point attack can be described within the same framework using the Jacobian matrix. In this study, we demonstrate that the rectangular MinRank attack can also be described within this framework. Furthermore, by leveraging this framework, we extend the feasible target ranks of the rectangular MinRank attack and use this extended attack to analyze the security of UOV and its variants. In conclusion, we confirm that the currently proposed parameters for UOV, MAYO, QR-UOV, and SNOVA are resistant to this attack.

Keywords: Post-Quantum Cryptography · Multivariate Cryptography · UOV · Rectangular MinRank Attack · Singular Point

1 Introduction

The security of widely used cryptographic schemes, such as RSA and elliptic curve cryptography, relies on the hardness of the factorization problem and the discrete logarithm problem. However, it is known that these schemes become vulnerable with the advent of large-scale quantum computers. In response to this threat, the National Institute of Standards and Technology (NIST) launched a standardization project in 2016 for post-quantum cryptography, which is resistant to quantum attacks.

Multivariate public key cryptography has been actively studied as one of the candidates for post-quantum cryptography, with its security relying on the hardness of solving systems of multivariate quadratic equations. For example, the multivariate public key signature scheme Rainbow — proposed by Ding and Schmidt as a multilayer extension of UOV — advanced to the third-round finalists in the NIST standardization project. However, the parameters proposed for Rainbow in that were broken by the rectangular MinRank attack introduced by Beullens [2] in 2021.

In 2022, NIST announced the standardization of four cryptographic schemes — CRYSTALS-Kyber, CRYSTALS-Dilithium, FALCON, and SPHINCS+ — as the interim results of its project. Additionally, to promote diversity in digital signature schemes, NIST launched an additional project focused on signature algorithms. For multivariate public key cryptography, UOV [5] remained a candidate in the second round, along with schemes that use transformations other than multilayer constructions, such as MAYO [4], QR-UOV [8], and SNOVA [15].

It was noted in [7] that the rectangular MinRank attack introduced by Beullens [2] is applicable to schemes such as MAYO and QR-UOV. In fact, in the first round of the additional signature project, this attack succeeded in breaking the proposed parameters of VOX, a variant of UOV. Furthermore, Pébereau [14] recently reported that the Kipnis–Shamir attack can be interpreted as the computation of singular points defined by the public key. An algebraic model of the Kipnis–Shamir attack was also proposed.

In this study, we first demonstrate that the rectangular MinRank attack can be described within the same framework as the Kipnis–Shamir attack and the singular point attack. Next, by generalizing Pébereau's discussion, we derive a formula for the target rank r in the rectangular MinRank attack for a given set of UOV parameters. In this framework, Pébereau's singular point attack solves a rectangular MinRank problem with a target rank of $r = m - 1$, while the existing rectangular MinRank attack [7] targets $r = n - o$. Here, n, m, and o denote the number of variables, the number of polynomials, and the number of oil variables, respectively. We then analyze the security of UOV and its variants against this attack. As a result, we found that all existing parameter sets remain secure against this attack.

2 UOV: Unbalanced Oil and Vinegar

In this section, we define a UOV key pair in a universal form.

Definition 1 (UOV Key Pair). *Let $\mathbb{F}_q$ be a finite field of order q, $x := (x_1, \ldots, x_n)^\top$ a set of variables, and $\mathcal{P}(x) := (\mathcal{P}_1(x), \ldots, \mathcal{P}_m(x))^\top$ a sequence of quadratic forms over $\mathbb{F}_q$. Let $\mathcal{O}$ be an o-dimensional totally isotropic subspace*[5] *of $\mathcal{P}$. Then, $\mathcal{P}$ and $\mathcal{O}$ are referred to as the* public key *and* private key*, respectively, of* $\mathrm{UOV}(q, n, m, o)$.

The public and private keys of the UOV scheme [5] and its variants (MAYO [4], QR-UOV [8], SNOVA [15]) can all be reduced to this structure. Therefore,

[5] Let K be a field, $x := (x_1, \ldots, x_n)^\top$ be a set of variables, and let $p(x) \in K[x]$ be a quadratic form. A linear subspace $\mathcal{O}$ of K^n is called a totally isotropic subspace of p if every element $a \in \mathcal{O}$ satisfies $p(a) = 0$. Furthermore, for a sequence of quadratic forms $\mathcal{P}$, if all entries of $\mathcal{P}$ have a totally isotropic subspace $\mathcal{O}$ in common, then $\mathcal{O}$ is called a totally isotropic subspace of $\mathcal{P}$.

to facilitate discussion independent of procedural or structural differences, we adopt this formulation in defining UOV.

Definition 2 (Central Map). *Let $\mathbb{F}_q$ be a finite field of order q, $x := (x_1, \dots, x_n)^\top$ a set of variables, and let $f(x) \in \mathbb{F}_q[x]$. If $f(x)$ is a quadratic form of the following structure, it is said to be in* UOV form*:*

$$f(x) = \sum_{\substack{1 \leq i \leq n \\ 1 \leq j \leq n-o}} c_{ij} x_i x_j.$$

Furthermore, a sequence of UOV *forms $\mathcal{F} := (\mathcal{F}_1, \dots, \mathcal{F}_m)^\top$ is called the* central map*, and a class of such quadratic form sequences is denoted by* $\mathrm{CM}(q, n, m, o)$*.*

Let $f(x)$ be a quadratic form. The bilinear form $f^*(x, y) := f(x+y) - f(x) - f(y)$ is called the polar form of f, and its representation matrix (a symmetric matrix) is denoted by f^*. Furthermore, for a sequence of quadratic forms $\mathcal{F}(x) = (\mathcal{F}_1(x), \dots, \mathcal{F}_m(x))^\top$, a sequence of the representation matrices of their polar forms is defined as $F^* := (F_1^*, \dots, F_m^*)^\top$.

A quadratic form $f(x)$ in UOV form does not contain terms involving only $x_{n-o+1}, \dots, x_n$. The representation matrix f^* of the polar form takes the following form:

$$f^* = \begin{pmatrix} *_{(n-o)\times(n-o)} & *_{(n-o)\times o} \\ *_{o\times(n-o)} & 0_{o\times o} \end{pmatrix}.$$

Traditionally, the central map $\mathcal{F}$ and a nonsingular matrix S are first generated uniformly at random, and the public key $\mathcal{P}$ is then constructed as $\mathcal{P} := \mathcal{F} \circ S$. The following equivalence holds for the public key.

Proposition 3. *For a sequence of quadratic forms $\mathcal{P}$, the following are equivalent:*

A. *There exists an $\mathcal{F} \in \mathrm{CM}(q, n, m, o)$ that is congruent*[6] *to $\mathcal{P}$.*
B. *$\mathcal{P}$ is a public key of* $\mathrm{UOV}(q, n, m, o)$*.*

For the private key, there are two representations: one using the variable transformation S and the other using the totally isotropic subspace $\mathcal{O}$. The following correspondence exists between these representations.

Let the basis of $\mathcal{O}$ be $b_1, \dots, b_o$, and assume that it is reduced to the form

$$(b_1, \dots, b_o) = \begin{pmatrix} B_{(n-o)\times o} \\ I_{o\times o} \end{pmatrix},$$

where $I_{o\times o}$ is the o-by-o identity matrix. In this case, the nonsingular matrix S is defined as

[6] Two sequences of quadratic forms $\mathcal{P}$ and $\mathcal{F}$ are said to be congruent if there exists a nonsingular variable transformation S such that $\mathcal{P} = \mathcal{F} \circ S$.

$$S = \begin{pmatrix} I_{(n-o)\times(n-o)} & B_{(n-o)\times o} \\ 0_{o\times(n-o)} & I_{o\times o} \end{pmatrix}.$$

It is known that restricting S in this manner does not compromise security [13].

3 Key Recovery Attacks against UOV

Let $\mathcal{P}$ and $\mathcal{O}$ be the public key and private key of $\mathrm{UOV}(q, n, m, o)$, respectively. An algorithm that takes $\mathcal{P}$ as input and outputs $\mathcal{O}$ is called a key recovery attack against $\mathrm{UOV}(q, n, m, o)$. In this section, we describe several existing key recovery attacks against UOV.

The following results are known regarding key recovery attacks against UOV.

Theorem 4 (Pébereau's One Vector Method [12])**.** *For* $\mathrm{UOV}(q, n, m, m)$*, if* α *linearly independent elements of* $\mathcal{O}$ *are obtained such that* $n - \alpha m \leq 2m$*, then the private key can be recovered in* $O(mn^{\omega})$ *time. Here,* ω *denotes the matrix multiplication exponent.*

For the currently proposed parameters of the UOV scheme, the value of α in Theorem 4 is 1.

Moreover, for parameter sets with $m \neq o$ (for example, in MAYO), a method for recovering the entire subspace $\mathcal{O}$ from a single vector $x \in \mathcal{O}$ has been proposed in Beullens' reconciliation attack against MAYO [3].

Elements of $\mathcal{O}$ are called secret vectors. Since these methods are efficient, the difficulty of finding α secret vectors essentially determines the difficulty of key recovery for UOV.

3.1 Rectangular MinRank Attack

One of the key recovery attacks against UOV is the rectangular MinRank attack [2]. The rectangular MinRank attack reduces the key recovery problem to a MinRank problem defined as follows.

Problem 5 (MinRank Problem)**.** *Let* $M = (M_1, \ldots, M_k)$ *be a sequence of* k *matrices of size* $n_{\mathrm{rows}} \times n_{\mathrm{cols}}$*, with* $n_{\mathrm{rows}} \geq n_{\mathrm{cols}}$*. Let* r *be an integer satisfying* $r < n_{\mathrm{cols}}$*. The* MinRank problem *is the problem of finding nontrivial coefficients* $c_1, \ldots, c_k$ *such that*

$$\mathrm{rank}(c_1 M_1 + \cdots + c_k M_k) \leq r.$$

The value r *is called the* target rank.

We define the following transformation, which gives rise to the term "rectangular".

Definition 6 (Rectangular Transformation [2, 10]). *Let $M = (M_1, \ldots, M_m)^\top$ be a sequence of m matrices, each of size $n \times n$. The column-wise and row-wise representations of each M_i $(i = 1, \ldots, m)$ are given as follows:*

$$M_i := \left(M_i^{\mathrm{col}(1)}, \cdots, M_i^{\mathrm{col}(n)}\right) = \begin{pmatrix} M_i^{\mathrm{row}(1)} \\ \vdots \\ M_i^{\mathrm{row}(n)} \end{pmatrix}.$$

Here, $M_i^{\mathrm{col}(j)}$ denotes the j-th column of M_i, and $M_i^{\mathrm{row}(j)}$ denotes the j-th row of M_i. We define the operations $\mathrm{Rect}_{\mathrm{col}}$ and $\mathrm{Rect}_{\mathrm{row}}$ on M as follows:

$$\mathrm{Rect}_{\mathrm{col}}(M) := \left(\left(M_1^{\mathrm{col}(1)}, \cdots, M_m^{\mathrm{col}(1)}\right) \cdots \left(M_1^{\mathrm{col}(n)}, \cdots, M_m^{\mathrm{col}(n)}\right)\right),$$

$$\mathrm{Rect}_{\mathrm{row}}(M) := \left(\left(M_1^{\mathrm{row}(1)\top}, \cdots, M_m^{\mathrm{row}(1)\top}\right) \cdots \left(M_1^{\mathrm{row}(n)\top}, \cdots, M_m^{\mathrm{row}(n)\top}\right)\right)^\top.$$

The following is clear from the definition.

Proposition 7. *Let M be a sequence of symmetric matrices . Then, the following holds:*

$$\mathrm{Rect}_{\mathrm{row}}(M) = \mathrm{Rect}_{\mathrm{col}}(M)^\top.$$

The rectangular MinRank attack is based on the following lemma.

Lemma 8 ([10]). *Let $\mathcal{P}$ and $\mathcal{O}$ be the public and private keys of $\mathrm{UOV}(q, n, m, o)$, respectively. Let $\mathcal{F}$ be an element of $\mathrm{CM}(q, n, m, o)$ congruent to $\mathcal{P}$, and let $\mathcal{P} = \mathcal{F} \circ S$. Define the rectangular transformation applied to the representation matrices of the polar forms of $\mathcal{P}$ and $\mathcal{F}$ as follows:*

$$(\tilde{P}_1, \ldots, \tilde{P}_n) := \mathrm{Rect}_{\mathrm{col}}(P^*)$$

$$(\tilde{F}_1, \ldots, \tilde{F}_n) := \mathrm{Rect}_{\mathrm{col}}(F^*)$$

Then, the following equation holds:

$$(\tilde{P}_1, \ldots, \tilde{P}_n) = (S^\top \tilde{F}_1, \ldots, S^\top \tilde{F}_n)S.$$

From the lemma, if $x = S^{-1}(0, \ldots, 0, a_{n-o+1}, \ldots, a_n)^\top \in \mathcal{O}$, then the following equation holds:

$$(\tilde{P}_1, \ldots, \tilde{P}_n) \cdot x = (S^\top \tilde{F}_{n-o+1}, \ldots, S^\top \tilde{F}_n) \cdot (a_{n-o+1}, \ldots, a_n)^\top.$$

For $1 \leq i \leq m$, since F_i^* has the form

$$F_i^* = \begin{pmatrix} *_{(n-o)\times(n-o)} & *_{(n-o)\times o} \\ *_{o\times(n-o)} & 0_{o\times o} \end{pmatrix},$$

it follows that for $n - o + 1 \leq j \leq n$,

$$\tilde{F}_j = \begin{pmatrix} *_{(n-o)\times m} \\ 0_{o\times m} \end{pmatrix}.$$

Therefore, when $n-o<m$, the rank of the linear combination $(\tilde{P}_1,\dots,\tilde{P}_n)\cdot x$ is at most $n-o$. This is equivalent to a MinRank problem with input $(\tilde{P}_1,\dots,\tilde{P}_n)$ and target rank $n-o$.

Since $x \in \mathcal{O}$, it also satisfies $\mathcal{P}(x)=0$. In the rectangular MinRank attack, the goal is to solve the following equations.

$$\begin{cases} \mathcal{P}(x)=0 \\ \mathrm{rank}(\mathrm{Rect}_{\mathrm{col}}(P^*)\cdot x) \le n-o \end{cases}$$

By Lemma 8, if $x \in \mathcal{O}$, then $\mathrm{rank}(\mathrm{Rect}_{\mathrm{col}}(P^*)\cdot x) \le n-o$. Thus, the solution space has at least dimension o. Therefore, even if $o-1$ variables are fixed to zero, a nontrivial solution still exists.

Since each element of $\mathrm{Rect}_{\mathrm{col}}(P^*)$ is an $n\times m$ matrix, the rectangular MinRank attack with target rank $n-o$ is not applicable to parameter sets where $\mathrm{Min}(n,m) \le n-o$. When these schemes are reduced to UOV, both MAYO and QR-UOV have $\mathrm{Min}(n,m) > n-o$. An analysis of these schemes under the rectangular MinRank attack is given in [7].

3.2 Support Minors Method

One of the algorithms for solving the MinRank problem is the support minors method [1]. This method reduces the MinRank problem to a system of bilinear equations as follows.

Let $M=(M_1,\dots,M_k)$ be the input of the MinRank problem, r be the target rank, and $x=(x_1,\dots,x_k)^\top$ be the solution. Consider the matrix $M(x) := x_1M_1+\cdots+x_kM_k$, whose entries are linear polynomials in x.

Let C be an arbitrary $r\times n_{\mathrm{cols}}$ submatrix of $M(x)$. For each $i=1,\dots,n_{\mathrm{rows}}$, define

$$Q_i(x) := \begin{pmatrix} C \\ M(x)^{\mathrm{row}(i)} \end{pmatrix}.$$

Denote the r-minor of C by $y_1,\dots,y_{\binom{n_{\mathrm{cols}}}{r}}$, and let the $(r+1)$-minor of $Q_i(x)$ be

$$Q_i^1(x,y),\dots,Q_i^{\binom{n_{\mathrm{cols}}}{r+1}}(x,y).$$

By cofactor expansion, each $Q_i^j(x,y)$ is expressed as a bilinear form in x and y. When the rank of $M(x)$ is at most r, the following system of equations holds:

$$Q_i^j(x,y)=0, \quad \left(1\le i\le n_{\mathrm{rows}}, 1\le j\le \binom{n_{\mathrm{cols}}}{r+1}\right).$$

In the support minors method, this system of equations—consisting of $k+\binom{n_{\mathrm{cols}}}{r}$ variables and $n_{\mathrm{rows}}\cdot\binom{n_{\mathrm{cols}}}{r+1}$ quadratic multivariate polynomials—is solved using the Wiedemann XL algorithm [6].

The complexity of the support minors method is estimated in [1] as

$$3\mathcal{M}(b_{\min},1)^2_{\text{cols}}(r+1)k.$$

Here, $\mathcal{M}(b,1)_{\text{cols}}$ denotes the number of columns in the Macaulay matrix $\mathcal{M}(b,1)$ of bidegree $(b,1)$. That is, for two sets of variables, $x=(1,\dots,x_k)^\top$ and $y=(1,\dots,y_{\binom{n_{\text{cols}}}{r}})^\top$, the value $\mathcal{M}(b,1)_{\text{cols}}$ represents the total number of monomials where the degree in x is b and the degree in y is 1.

The smallest value $b_{\min}$ satisfying $\mathcal{R}(b) > \mathcal{M}(b,1)_{\text{cols}} - 1$ is considered to yield the most efficient solution to the system of equations $\{Q_i^j(x,y) = 0\}_{(1\le i\le n_{\text{rows}},1\le j\le \binom{n_{\text{cols}}}{r+1})}$. Here, $\mathcal{R}(b)$ corresponds to the rank of $\mathcal{M}(b,1)$, which is expected to be estimated well-approximated by the following expression:

$$\mathcal{R}(b) := \sum_{i=1}^{b}(-1)^{i+1}\binom{n_{\text{cols}}}{r+i}\binom{n_{\text{rows}}+i-1}{i}\binom{r+b-i-1}{b-i}.$$

In this paper, we focus on the MinRank problem with $\{\mathcal{P}_i(x)=0\}_{1\le i\le m}$. In this case, for the rank $\widetilde{\mathcal{R}}(b)$ of the corresponding Macaulay matrix $\widetilde{\mathcal{M}}(b,1)$, we take $b_{\min}$ as the smallest value b satisfying $\widetilde{\mathcal{R}}(b) > \mathcal{M}(b,1)_{\text{cols}} - 1$. Note that $\widetilde{\mathcal{M}}(b,1)_{\text{cols}} = \mathcal{M}(b,1)_{\text{cols}}$. The value $\mathcal{M}(b,1)_{\text{col}} - \widetilde{\mathcal{R}}(b)$ is often well-estimated the coefficient of t^b in $(1-t^2)^m \cdot \sum_{i\ge 0}(\mathcal{M}(i,1)_{\text{cols}} - \mathcal{R}(i))t^i$ under the assumption in [2].

3.3 Kipnis–Shamir Attack

Another key recovery attack against UOV is the Kipnis–Shamir attack, which is based on the following lemma.

Lemma 9 ([3]). *Let $\mathcal{P}$ and $\mathcal{O}$ be the public and private keys of* UOV(q,n,m,o), *respectively. Let M be the set of all nonsingular linear combinations of P^*. Then, for any $M_i \in M$, the following holds:*

$$M_i\mathcal{O} \subset \mathcal{O}^\perp.$$

From this lemma, defining $M_{ij} := M_i^{-1}M_j$ for $M_i, M_j \in M$, we see that $\mathcal{O}$ is an invariant subspace of M_{ij} when $n = 2o$. In the Kipnis–Shamir attack, the eigenspace of M_{ij} is computed as its invariant subspace.

Furthermore, by accounting for the one vector method, the Kipnis–Shamir attack requires computing only a single eigenvector of M_{ij}.

3.4 Singular Point Attack

Another key recovery attack against UOV is the singular point attack [14]. It is described as follows.

Definition 10 (Jacobian Matrix of Quadratic Forms [14]). *Let $x := (x_1, \ldots, x_n)^\top$ be a set of variables, $\mathcal{P}(x) = (\mathcal{P}_1(x), \ldots, \mathcal{P}_m(x))^\mathsf{T}$ a sequence of quadratic forms, and let $P^* = (P_1^*, \ldots, P_m^*)^\mathsf{T}$ the sequence of their representation matrices of polar forms. The* Jacobian matrix *of $\mathcal{P}$, denoted by $\mathrm{Jac}_P(x)$, is defined as follows:*

$$\mathrm{Jac}_{\mathcal{P}}(x) := \begin{pmatrix} x^\mathsf{T} \cdot P_1^* \\ \vdots \\ x^\mathsf{T} \cdot P_m^* \end{pmatrix}.$$

This is an $m \times n$ matrix whose elements are linear polynomials.

This definition is consistent with the standard definition using partial derivatives.

Lemma 11 (Equivalence of Definitions of the Jacobian Matrix [14]). *Let $x := (x_1, \ldots, x_n)^\top$ be a set of variables, $\mathcal{P}(x) = (\mathcal{P}_1(x), \ldots, \mathcal{P}_m(x))^\top$ a sequence of quadratic forms, and let $P^* = (P_1^*, \ldots, P_m^*)^\top$ the sequence of their representation matrices of polar forms. Then, the following equation holds:*

$$\begin{pmatrix} \frac{\partial \mathcal{P}_1}{\partial x_1} & \cdots & \frac{\partial \mathcal{P}_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial \mathcal{P}_m}{\partial x_1} & \cdots & \frac{\partial \mathcal{P}_m}{\partial x_n} \end{pmatrix} = \begin{pmatrix} x^\top \cdot P_1^* \\ \vdots \\ x^\top \cdot P_m^* \end{pmatrix}.$$

For a sequence of polynomials $\mathcal{P} = (\mathcal{P}_1, \ldots, \mathcal{P}_m)^\top$, the set of all x satisfying $\mathcal{P}_1(x) = \cdots = \mathcal{P}_m(x) = 0$ is called the affine variety of $\mathcal{P}$ and is denoted by $\mathbb{V}(\mathcal{P})$. The dimension[7] of the affine variety $\mathbb{V}(\mathcal{P})$ is written as $\dim \mathbb{V}(\mathcal{P})$.

We introduce the following hypothesis regarding the dimension of the affine variety defined by the public key of the UOV.

Hypothesis 12. *Let $\mathcal{P}$ and $\mathcal{O}$ be the public and private keys of* $\mathrm{UOV}(q, n, m, o)$*, respectively. Then, the following holds:*

$$\dim \mathbb{V}(\mathcal{P}) = \max(n - m, o).$$

When m polynomials in n variables are generated uniformly at random, the dimension of their solution space is generally $\max(n-m, -1)$. Here, a dimension of -1 indicates that the system of equations has no solution. Given this fact and that the UOV public key has an o-dimensional totally isotropic subspace, the above hypothesis is considered reasonable.

Definition 13 (Singular Point). *Let $\mathcal{P}$ be a sequence of quadratic forms, and let $\mathbb{V}(\mathcal{P})$ be the affine variety defined by $\mathcal{P}$. A point $a \in \mathbb{V}(\mathcal{P})$ is called a* singular point *of $\mathbb{V}(\mathcal{P})$ if it satisfies*

$$\mathrm{rank}(\mathrm{Jac}_{\mathcal{P}}(a)) < n - \dim \mathbb{V}(\mathcal{P}).$$

[7] For a formal definition of the dimension of a variety, see, for example, Chapter 9 of "Cox, David A., John B. Little, and Donal O'Shea. Ideals, Varieties, and Algorithms: An Introduction to Computational Algebraic Geometry and Commutative Algebra. 4th ed. Springer. 2007".

The set of all singular points of $\mathbb{V}(\mathcal{P})$ *is called the* singular locus *and is denoted by* Σ*.*

Under Hypothesis 12, for a public key $\mathcal{P}$ of $\mathrm{UOV}(q,n,m,o)$, a point $a \in \mathbb{V}(\mathcal{P})$ is a singular point if it satisfies

$$\mathrm{rank}(\mathrm{Jac}_{\mathcal{P}}(a)) < n - \max(n-m, o).$$

Regarding singular points of the UOV, the following hypothesis was introduced in [14].

Hypothesis 14 (Kipnis–Patarin–Goubin)**.** *Let* K *be an algebraically closed field, and let* $x := (x_1, \dots, x_n)^\top$ *be a set of variables. Suppose that a sequence of quadratic forms* $\mathcal{P}$ *over* $K[x]$ *has a totally isotropic subspace* $\mathcal{O}$*. Then, for the singular locus* Σ *of* $\mathbb{V}(\mathcal{P})$*, the following holds:*

$$\Sigma \subset \mathcal{O}.$$

Hypothesis 14 is based on the assumption that the field is algebraically closed. Therefore, it does not strictly hold over finite fields.

If $\Sigma \subset \mathcal{O}$ holds, then using those singular points in the one vector method enables recovery of the private key. This key recovery attack is referred to as the singular point attack. If $\Sigma \subset \mathcal{O}$ does not strictly hold, then a singular point obtained uniformly at random belongs to $\mathcal{O}$ with probability

$$\frac{|\Sigma \cap \mathcal{O}|}{|\Sigma|}.$$

The following theorem addresses the dimension of the singular locus of UOV.

Theorem 15 (Pébereau [14])**.** *Let* $\mathcal{P}$ *and* $\mathcal{O}$ *be the public and private keys of* $\mathrm{UOV}(q,n,m,o)$*, respectively. Suppose that* $n - m \geq o$*, and let* Σ *be the singular locus of* $\mathbb{V}(\mathcal{P})$*. Then, the following inequality holds:*

$$\dim(\Sigma \cap \mathcal{O}) \geq 2o + m - n - 1.$$

For computing singular points, [14] proposes the following modeling, called bihomogeneous modeling.

Assume $n - m \geq o$. Since the condition that the Jacobian matrix is not full-rank is equivalent to the existence of a nontrivial kernel, it suffices to solve the following equation[8].

$$\begin{cases} x \in \mathbb{F}_q^n, y \in \mathbb{F}_q^m, x \neq 0, y \neq 0 \\ \mathcal{P}(x) = 0 \\ y^\intercal \mathrm{Jac}_{\mathcal{P}}(x) = 0 \end{cases}$$

The following relationship between the (one vector) Kipnis–Shamir attack and singular point attack is known.

[8] In practice, [14] proposes a method that uses only a subset of the rows of $\mathrm{Jac}_{\mathcal{P}}(x)$

Theorem 16 ([14]). *Let $\mathcal{P}$ be the public key of* $\mathrm{UOV}(q, n, m, o)$*. Then, if x is an eigenvector of M_{ij}, the Jacobian matrix* $\mathrm{Jac}_{\mathcal{P}}(x)$ *is not full rank.*

Thus, both the Kipnis–Shamir attack and the singular point attack can be described within a common framework based on the rank of the Jacobian matrix. We refer to this framework as the Jacobian framework.

4 Extended Rectangular MinRank Attack

In Section 4.1, we show that the rectangular MinRank attack can be described within the Jacobian framework, and we further generalize Theorem 15. In Section 4.2, we experimentally examine the extent to which Hypothesis 14 holds over finite fields and for singular points of lower rank. In Section 4.3, based on these results, we propose the extended rectangular MinRank attack, an attack with an extended target rank.

4.1 Rectangular MinRank Attack in the Jacobian Framework

In summary, the rectangular MinRank attack solves the following system of equations:

$$\begin{cases} \mathcal{P}(x) = 0 \\ \mathrm{rank}(\mathrm{Rect}_{\mathrm{col}}(P^*) \cdot x) \leq n - o \end{cases}$$

while the singular point attack solves:

$$\begin{cases} \mathcal{P}(x) = 0 \\ \mathrm{rank}(\mathrm{Jac}_{\mathcal{P}}(x)) < n - \dim \mathbb{V}(\mathcal{P}) \end{cases}$$

Both attacks aim to find points on $\mathbb{V}(\mathcal{P})$ where a matrix determined by $\mathcal{P}$ and x has its rank bounded by some value. Thus, both attacks share a similar structure.

The following relationship holds between the two attacks.

Theorem 17. *Let $\mathcal{P}$ be a sequence of quadratic forms. Then, the following holds:*

$$\mathrm{rank}(\mathrm{Jac}_{\mathcal{P}}(x)) = \mathrm{rank}(\mathrm{Rect}_{\mathrm{col}}(P^*) \cdot x).$$

Proof. Let $\mathcal{P}$ be a sequence of quadratic forms, and let $P^* = (P_1^*, \ldots, P_m^*)^\top$ be the sequence of representation matrices of their polar forms. Then,

$$\mathrm{Jac}_{\mathcal{P}}(x) := \begin{pmatrix} (x_1, \cdots, x_n) \begin{pmatrix} P_1^{*\,\mathrm{row}(1)} \\ \vdots \\ P_1^{*\,\mathrm{row}(n)} \end{pmatrix} \\ \vdots \\ (x_1, \cdots, x_n) \begin{pmatrix} P_m^{*\,\mathrm{row}(1)} \\ \vdots \\ P_m^{*\,\mathrm{row}(n)} \end{pmatrix} \end{pmatrix}$$

$$
\begin{aligned}
&= \begin{pmatrix} x_1 \cdot P_1^{*\,\mathrm{row}(1)} + \cdots + x_n \cdot P_1^{*\,\mathrm{row}(n)} \\ \vdots \\ x_1 \cdot P_m^{*\,\mathrm{row}(1)} + \cdots + x_n \cdot P_m^{*\,\mathrm{row}(n)} \end{pmatrix} \\
&= x_1 \cdot \begin{pmatrix} P_1^{*\,\mathrm{row}(1)} \\ \vdots \\ P_m^{*\,\mathrm{row}(1)} \end{pmatrix} + \cdots + x_n \cdot \begin{pmatrix} P_1^{*\,\mathrm{row}(n)} \\ \vdots \\ P_m^{*\,\mathrm{row}(n)} \end{pmatrix} \\
&= x^\top \cdot \mathrm{Rect}_{\mathrm{row}}(P^*).
\end{aligned}
$$

Since each element of P^* is a symmetric matrix, taking the transpose of both sides gives:

$$\mathrm{Jac}_{\mathcal{P}}(x)^\top = \mathrm{Rect}_{\mathrm{col}}(P^*) \cdot x.$$

Since the rank of a matrix is invariant under transposition, the rank of this matrix is equal to that of $\mathrm{Jac}_{\mathcal{P}}(x)$. □

We also generalize Theorem 15 to the case of $\dim \mathbb{V}(\mathcal{P}) = o$, i.e., $n - m \leq o$, so that it can be applied to schemes such as MAYO and QR-UOV.

Theorem 18. *Let $\mathcal{P}$ and $\mathcal{O}$ be the public and private keys of* $\mathrm{UOV}(q, n, m, o)$, *respectively. Assume $r < \min(n - o, m)$. Define the set Σ_r by*

$$\Sigma_r := \{x \mid \mathcal{P}(x) = 0 \wedge \mathrm{rank}(\mathrm{Jac}_{\mathcal{P}}(x)) \leq r\}.$$

Then, the following inequality holds:

$$\dim(\Sigma_r \cap \mathcal{O}) \geq o - (m - r)((n - o) - r).$$

The proof can be found in [14]. This theorem is a natural extension of Theorem 15. In fact, by assuming $n - m \geq o$ and substituting $r = m - 1$, we obtain the same result as in Theorem 15.

Based on Theorem 18, we introduce the following notation.

Definition 19. *Let n, m, o, r be natural numbers. Define:*

$$
\begin{aligned}
D_{n,m,o}(r) &:= o - (m - r)(n - o - r), \\
R(n, m, o) &:= \{r \in \mathbb{N} \mid r < \min(n - o, m) \wedge D_{n,m,o}(r) > 0\}, \\
R^+(n, m, o) &:= \begin{cases} R(n, m, o) \cup \{n - o\} & (n - m < o) \\ R(n, m, o) & (n - m \geq o) \end{cases}
\end{aligned}
$$

$D_{n,m,o}(r)$ is often abbreviated as $D(r)$. $R^+(n, m, o)$ is the set of positive integers expected to be feasible as target ranks for the attack proposed later. In addition to singular points, when $n - m < o$, the rank $n - o$ is also included as a target rank as in the original rectangular MinRank attack.

4.2 Number and Location of Singular Points of UOV

If $r \in R(n,m,o)$, can we conclude that there exists a linear combination of the transformed public key matrices with rank r? Additionally, Hypothesis 14 states that over an algebraically closed field, all singular points of $\mathbb{V}(\mathcal{P})$ are contained in $\mathcal{O}$. We ask: to what extent does this assumption hold over finite fields? In this section, we experimentally investigate these questions.

For parameter sets within a range where exhaustive search of $\mathbb{V}(\mathcal{P})$ is feasible, we generated 100 UOV key pairs (using 10 different random seeds each for the public and private keys). We then measured the following:

A Average of $|\Sigma_r|$ over all seeds.
(In parentheses: proportion of seeds for which $\Sigma_r \neq \{0\}$ out of all seeds.)
B Average of $|\Sigma_r \cap \mathcal{O}|$ over all seeds.
(In parentheses: proportion of seeds for which $\Sigma_r \cap \mathcal{O} \neq \{0\}$ out of all seeds.)
C Average of $\frac{|\Sigma_r \cap \mathcal{O}|}{|\Sigma_r|}$ over seeds for which $\Sigma_r \neq \{0\}$.

The results are presented in Table 1.

Table 1. Experiments for Σ_r

No.	q,n,m,o	r	$D(r)$	A	B	C
1	$2,12,10,6$	5	1	5 (99%)	5 (99%)	100%
2	$16,12,10,6$	5	1	18 (72%)	18 (72%)	100%
3	$31,12,10,6$	5	1	34 (61%)	34 (61%)	100%
4	$31,9,3,5$	2	3	30694 (100%)	30693 (100%)	100%
5	$31,10,3,5$	2	2	1002 (100%)	1001 (100%)	100%
6	$31,12,6,6$	5	5	29557623 (100%)	29557623 (100%)	100%
		4	2	990 (100%)	990 (100%)	100%
7	$31,8,4,3$	3	1	28 (64%)	27 (62%)	95%

Each value is rounded to the nearest integer at the first decimal place.

Rows 1, 2, and 3 correspond to cases where n,m,o are fixed while the field size increases. In particular, rows 1 and 2 vary only the field size while keeping the characteristic fixed. Rows 1, 2, and 3 satisfy $\dim \mathbb{V}(\mathcal{P}) = o$, and rows 4, 5, 6, and 7 satisfy $\dim \mathbb{V}(\mathcal{P}) = n - m$. Rows 4 and 5 correspond to cases where n is varied while keeping other parameters fixed, and row 6 represents a parameter set with multiple singular points of different ranks. Row 7 presents an example in which $\frac{|\Sigma_r \cap \mathcal{O}|}{|\Sigma_r|}$ does not reach 100%.

Despite differences in these parameter sets, the following observations hold consistently:

- $|\Sigma_r \cap \mathcal{O}|$ is approximately $q^{D(r)}$.

– $\frac{|\Sigma_r \cap \mathcal{O}|}{|\Sigma_r|}$ is nearly 100% (whenever $\Sigma_r \neq \{0\}$).

The first result supports Theorem 18, and the second supports relevalence of Hypothesis 14 over finite fields.

In the Table 1, the ratio $\frac{|\Sigma_r \cap \mathcal{O}|}{|\Sigma_r|}$ appears as 100% for almost all cases. However, this is due to rounding to the nearest integer at the first decimal place, and some parameter sets do show slight deviations. Therefore, Hypothesis 14 does not strictly hold over finite fields. However, since Σ_r and $\mathcal{O}$ intersect with extremely high probability, this discrepancy from the statement in Hypothesis 14 is not considered significant for cryptographic attacks.

For $D(r) = 1$, a non-negligible fraction of seeds yield $\Sigma_r \cap \mathcal{O} = \{0\}$. This reduces the success probability of the attack proposed later. Moreover, according to the experimental results, all parameter sets for which $\frac{|\Sigma_r \cap \mathcal{O}|}{|\Sigma_r|}$ does not reach 100%—as in Row 7—correspond to the case $D(r) = 1$. However, based on later complexity estimates, the proposed attack is more efficient with the target rank r where $D(r)$ is large. From this perspective, cases where $D(r) = 1$ are not particularly important.

In conclusion, for any $r \in R(n, m, o)$, almost every point x with $x \in \Sigma_r$ can be assumed to also lie in $\mathcal{O}$. Furthermore, on average, there exist $q^{D(r)}$ such points. Although Table 1 shows only a few parameter sets, we have confirmed that these findings hold for all parameter choices that were feasible to test.

4.3 Extended Rectangular MinRank Attack

Based on the discussion in the previous section, the target rank of the rectangular MinRank attack can be extended as follows.

Heuristic 20 (Extended Rectangular MinRank Attack)**.** *Let $\mathcal{P}$ and $\mathcal{O}$ be the public and private keys of* UOV(q, n, m, o)*, respectively, and let $r \in R^+(n, m, o)$. Then, if x is a solution to the following equations, we consider x to lie in $\mathcal{O}$:*

$$\begin{cases} \mathcal{P}(x) = 0 \\ \operatorname{rank}(\mathrm{Rect}_{\mathrm{col}}(P^*) \cdot x) \leq r \end{cases}$$

For example, in the case $(n, m, o) = (66, 64, 8)$, solving $D_{66,64,8}(r) > 0$ yields

$$8 - (64 - r)(58 - r) > 0,$$

which in turn implies

$$57 \leq r \leq 65$$

Thus,

$$R^+(66, 64, 8) = \{57, 58\}.$$

This result shows that the target rank can be set to a value smaller than the previously established target rank $r = n - o = 58$ in the original rectangular MinRank attack.

Similar to the original rectangular MinRank attack, we first reduce the equation $\mathrm{rank}(\mathrm{Rect}_{\mathrm{col}}(P^*) \cdot x) \leq r$ to a system of quadratic multivariate polynomial equations using the support minors modeling. We then solve this system, together with $\mathcal{P}(x) = 0$, using the Wiedemann XL algorithm.

The solution space of this system has dimension $D(r)$. Therefore, we expect that a nontrivial solution can still be obtained even if $D(r) - 1$ variables are fixed to zero.

5 Security Analysis of UOV and Its Variants

In this section, we discuss the complexity estimation of the extended rectangular MinRank attack against UOV [5], MAYO [4], QR-UOV [8], and SNOVA [15] in the NIST PQC standardization project additional signatures.

For details on complexity analysis of rectangular MinRank attack, refer to [7]. The subsequent complexity estimations are based on this approach (see Appendix B).

UOV Against the currently proposed UOV parameters, the extended rectangular MinRank attack can only be mounted with target rank $r = m - 1$. In this case, our attack coincides with Pébereau's singular point attack using the bihomogeneous modeling.

Table 2 in Appendix A presents the complexity estimation of the extended rectangular MinRank attack against UOV[9]. As a result, we found that all parameter sets meet the required security level against this attack.

MAYO It was shown in [7] that the round-1 parameters of MAYO were applicable of the original rectangular MinRank attack. Consequently, in round 2, those parameters were modified to prevent the applicability of the original rectangular MinRank attack. However, our extended rectangular MinRank attack still applies to the round-2 parameters.

MAYO's parameters (q, n, m, o, k) can be directly mapped to $\mathrm{UOV}(q, n, m, o)$. Table 3 in Appedix A presents the complexity estimation of the extended rectangular MinRank attack against MAYO[10].

As a result, we found the following:

- A larger target rank r leads to greater efficiency of the attack.
- All parameter sets meet the required security level against this attack.

QR-UOV Among the known reductions of a QR-UOV key pair to a UOV key pair, only one yields UOV key pair where the extended rectangular MinRank attack is applicable: converting the QR-UOV parameters (q, v, m, l) into

[9] https://www.uovsig.org/
[10] https://pqmayo.org/

UOV($q^l, (v+m)/l, m, m/l$). Under this transformation, Table 4 in Appendix A presents the complexity estimation of the extended rectangular MinRank attack against QR-UOV[11].

As a result, we found the following:

- A larger target rank r leads to greater efficiency of the attack.
- All parameter sets meet the required security level against this attack.

SNOVA SNOVA constructs a UOV scheme over a noncommutative matrix ring. In a key recovery context, it is known that a SNOVA scheme with parameters (v, o, q, l) can be regarded as UOV($q, l(o+v), l^2o, lo$) [9, 11].

In a key recovery attack against SNOVA, given the public key P_i ($i = 1, \ldots, l^2o$), the goal is to find matrices F and T such that

$$P_i = T^\top F_i T \quad (i = 1, \ldots, l^2o).$$

Here, T and F_i have the following forms:

$$T = \begin{pmatrix} I^{11} & T^{12} \\ 0 & I^{22} \end{pmatrix}, \quad F_i = \begin{pmatrix} F_i^{11} & F_i^{12} \\ F_i^{21} & 0 \end{pmatrix}.$$

Note that P_i is not necessarily symmetric. Therefore, by taking the transpose, we obtain $2l^2o$ relations[12].

Since P_i is not symmetric, Proposition 7 must be rewritten as follows.

Proposition 21. *Let $M = (M_1, \cdots, M_k)$ be a sequence of square matrices (not necessarily symmetric). Then, the following holds:*

$$\begin{aligned} &\mathrm{Rect}_{\mathrm{row}}(M_1, \cdots, M_k, M_1^\top, \cdots, M_k^\top) \\ =&\mathrm{Rect}_{\mathrm{col}}(M_1^\top, \cdots, M_k^\top, M_1, \cdots, M_k)^\top. \end{aligned}$$

Except for the fact that the width of the matrices forming the MinRank problem is doubled, the same argument applies to SNOVA as well.

Table 5 in Appendix A presents the complexity estimation of the extended rectangular MinRank attack against SNOVA[13].

As a result, we found the following:

- The only feasible target rank r for this attack is $n - o$.
- All parameter sets meet the required security level against this attack.

[11] https://info.isl.ntt.co.jp/crypt/qruov/

[12] Note that this refers to the conditions on matrices; the number of polynomial equations remains l^2o

[13] https://snova.pqclab.org/

6 Conclusion

In this study, we first demonstrated that the rectangular MinRank attack can be described within the Jacobian framework. As a result, three key recovery attacks against UOV—Kipnis–Shamir attack, singular point attack, and rectangular MinRank attack—were unified under a common theoretical framework.

Next, by generalizing Pébereau's theorem on the dimension of the singular locus of UOV, we extended the range of feasible target ranks for the rectangular MinRank attack. Based on this, we proposed the extended rectangular MinRank attack and evaluated the security of UOV and its variants against this attack. In conclusion, we found that all existing parameter sets remain secure against this attack. Additionally, it was observed that choosing a larger r as the target rank results in a more efficient attack. This is because a larger r corresponds to a larger $D(r)$. Since $D(r) - 1$ variables are fixed to zero during computation, a larger $D(r)$ reduces the number of variables in the system of equations to be solved.

A key challenge for future research is to clarify the positioning of other key recovery attacks against UOV within the Jacobian framework. Naturally, there may exist key recovery attacks that cannot be described using the Jacobian framework, in which case an appropriate extension of the framework may be necessary. If all existing key recovery attacks can be formulated within a unified framework, it will allow for a qualitative comparison of their efficiency, enabling the identification of the most effective key recovery attack. This, in turn, would facilitate the security analysis of newly proposed UOV parameters and variants, making their evaluation more systematic and efficient.

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A. Tables for Complexity

In the following tables estimating the complexity of the extended rectangular MinRank attack on multivariate signatures selected as the second round candidates for NIST PQC additional signatures:

- Complexity represents the gate complexity (denoted as $\log_2$), calculated based on the number of multiplications over the finite field $\mathbb{F}_q$ using the formula:

$$\sharp\text{gates} = \sharp\text{multiplications} \times (2(\log_2 q)^2 + \log_2 q).$$

  The values are rounded to the nearest integer at the first decimal place.
- In the MinRank problem, we assume that the input matrices are tall, i.e., the number of rows exceeds the number of columns.
- $D(r) - 1$ variables are fixed to zero.

– If q is even, the number of rows used is reduced by 1.[14]
– Values marked with "*" indicate that $b_{\min}$ could not be determined[15]; instead, the value was computed by setting $b_{\min} = r + 2$.

Table 2. Estimated gate count of the extended rectangular MinRank attack applied to UOV of NIST PQC standardization project additional signatures round 2

SL	UOV (q, n, m, o)	r	$D(r)$	Complexity	$b_{\min}$
1	uov-Ip $(256, 112, 44, 44)$	43	19	*275	*45
1	uov-Is $(16, 160, 64, 64)$	63	31	374	63
3	uov-III $(256, 184, 72, 72)$	71	31	*436	*73
5	uov-V $(256, 244, 96, 96)$	95	43	567	96

Table 3. Estimated gate count of the extended rectangular MinRank attack applied to MAYO of NIST PQC standardization project additional signatures round 2

SL	MAYO (q, n, m, o)	as UOV (q, n, m, o)	r	$D(r)$	Complexity	$b_{\min}$
1	MAYO$_1$ $(16, 86, 78, 8)$	$(16, 86, 78, 8)$	77	7	276	51
			76	4	304	56
1	MAYO$_2$ $(16, 81, 64, 17)$	$(16, 81, 64, 17)$	63	16	210	35
			62	13	224	35
			61	8	251	39
			60	1	286	45
3	MAYO$_3$ $(16, 118, 108, 10)$	$(16, 118, 108, 10)$	107	9	360	66
			106	6	393	72
			105	1	447	86
5	MAYO$_5$ $(16, 154, 142, 12)$	$(16, 154, 142, 12)$	141	11	459	84
			140	8	495	91
			139	3	555	107

[14] This is based on the results of [2]

[15] In the complexity analysis of the support minors method in Section 3.2, the theoretical framework allows complexity estimation only within the range $b < r + 2$. If $\widetilde{\mathcal{R}}(b) > \mathcal{M}(b, 1)_{\text{cols}} - 1$ does not hold for any b in this range, the complexity estimation fails. Note that this does not imply that the attack is infeasible.

Table 4. Estimated gate count of the extended rectangular MinRank attack applied to QR-UOV of NIST PQC standardization project additional signatures round 2

SL	QR-UOV (q, v, m, l)	as UOV (q, n, m, o)	r	$D(r)$	Complexity	$b_{\min}$
1	$(7, 740, 100, 10)$	$(7^{10}, 84, 100, 10)$	74	10	202	12
1	$(31, 165, 60, 3)$	$(31^3, 75, 60, 20)$	55	20	161	14
			54	14	185	17
			53	6	214	21
1	$(31, 600, 70, 10)$	$(31^{10}, 67, 70, 7)$	60	7	183	14
1	$(127, 156, 54, 3)$	$(127^3, 70, 54, 18)$	52	18	160	20
			51	15	175	21
			50	10	194	23
			49	3	226	28
3	$(7, 1100, 140, 10)$	$(7^{10}, 124, 140, 14)$	110	14	289	18
3	$(31, 246, 87, 3)$	$(31^3, 111, 87, 29)$	82	29	226	23
			81	23	251	26
			80	15	289	32
			79	5	331	39
3	$(31, 890, 100, 10)$	$31^{10}, 99, 100, 10$	89	10	258	20
3	$(127, 228, 78, 3)$	$(127^3, 102, 78, 26)$	76	26	219	29
			75	23	234	30
			74	18	259	33
			73	11	291	38
			72	2	332	45
5	$(7, 1490, 190, 10)$	$(7^{10}, 168, 190, 19)$	149	19	373	22
5	$(31, 324, 114, 3)$	$(31^3, 146, 114, 38)$	108	38	288	30
			107	31	318	34
			106	22	357	40
			105	11	405	48
5	$(31, 1120, 120, 10)$	$(31^{10}, 124, 120, 12)$	112	12	315	31
			111	3	363	39
5	$(127, 306, 105, 3)$	$(127^3, 137, 105, 35)$	102	35	277	35
			101	31	298	37
			100	25	331	42
			99	17	368	48
			98	7	414	56

Table 5. Estimated gate count of the extended rectangular MinRank attack applied to SNOVA of NIST PQC standardization project additional signatures round 2

SL	SNOVA (v,o,q,l)	as UOV (q,n,m,o)	r	$D(r)$	Complexity	$b_{\min}$
1	$(37,17,16,2)$	$(16,108,68,34)$	74	34	229	2
1	$(25,8,16,3)$	$(16,99,72,24)$	75	24	220	5
1	$(24,5,16,4)$	$(16,116,80,20)$	96	20	276	13
3	$(56,25,16,2)$	$(16,162,100,50)$	112	50	337	3
3	$(49,11,16,3)$	$(16,180,99,33)$	147	33	476	28
3	$(37,8,16,4)$	$(16,180,128,32)$	148	32	395	15
3	$(24,5,16,5)$	$(16,145,125,25)$	120	25	294	9
5	$(75,33,16,2)$	$(16,216,132,66)$	150	66	446	4
5	$(66,15,16,3)$	$(16,243,135,45)$	198	45	622	34
5	$(60,10,16,4)$	$(16,280,160,40)$	240	40	776	61
5	$(29,6,16,5)$	$(16,175,150,30)$	145	30	346	10

B. Validity of Complexity Analysis

To confirm that complexity analysis of [7] remains valid in the context of our attack, we conducted experiments on several small parameter sets to check whether the actual value of $b_{\min}$ matches the theoretical prediction. Here, the smallest feasible degree $b_{\min}$ and Macaulay matrix $\widetilde{\mathcal{M}}(b,1)$ are defined as in the complexity analysis of the support minors method in Section 3.2.

As a result, there were cases in which the actual value of $\mathrm{corank}(\widetilde{\mathcal{M}}(b,1))$ did not drop below 1, and thus a measurable value for $b_{\min}$ could not be obtained. However, as b increased, $\mathrm{corank}(\widetilde{\mathcal{M}}(b,1))$ converged to a constant value c.

The following may explain this phenomenon: Since Theorem 18 gives a lower bound, the actual dimension $\dim(\Sigma_r \cap \mathcal{O})$ does not necessarily equal $D(r)$; in fact, it may exceed $D(r)$. When this happens, the corank of $\widetilde{\mathcal{M}}(b,1)$ may remain greater than 1, meaning that more than $D(r)-1$ variables can be fixed while still obtaining a nontrivial solution. It should be noted that this phenomenon cannot occur in the original rectangular MinRank attack [7], since in that case, the dimension of the solution space arising from the modeling never exceeds $\dim \mathbb{V}(\mathcal{P}) = o$.

Let $b'_{\min}$ be the smallest value of b such that $\mathrm{corank}(\widetilde{\mathcal{M}}(b,1)) \leq c$. It was observed that this $b'_{\min}$ matched the theoretical value of $b_{\min}$ computed under the assumption that $\widetilde{\mathcal{R}}(b) > \mathcal{M}(b,1)_{\mathrm{cols}} - 1$ in most cases, with only slight deviations observed in a few samples.

In this sense, the complexity analysis provided in [7] is considered to be valid for our attack as well.

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